



THE ANSWERS MUST BE ATTEMPTED ON THE ANSWER SHEET PROVIDED

Q.1. Write short answer of the following questions: (10×3=30)

(i)- What is meant by operators and state vectors in quantum mechanics? Also explain dimensionality and basis.

†(ii)- What is meant by discrete and continuous spectrum? Explain it by considering the example of a general potential.

(iii)- State any three postulates of quantum mechanics.

(iv)- If x and p are position and momentum variables and

$$\hat{A} = \frac{1}{\sqrt{2m\hbar\omega}}(m\omega x + ip)$$

Show that $[\hat{A}, \hat{A}^\dagger] = 1$.

(v)- Find out classical amplitude for a particle undergoing simple harmonic motion if its energy is $\hbar\omega/2$.

• (vi)- A particle is in state given by wave function,

$$\psi(x, t) = e^{-ax^2 - bt}$$

where a and b are constants.

Is it a stationary state? Find expectation values of position and momentum operators.

✱(vii)- Show that if an operator $\hat{A}$ has an eigen value $\alpha \neq \alpha^*$, then $\hat{A}$ is not Hermitian.

✱(viii)- Given that in spherical polar coordinates $\hat{L}_z = -i\hbar \frac{\partial}{\partial \phi}$, show that

$$[\hat{L}_z, \sin\phi] = -i\hbar \cos\phi$$

✱(ix)- The operator for z-component of angular momentum is $\hat{L}_z = -i\hbar \frac{\partial}{\partial \phi}$. Determine whether or not $\sin m\phi$ is its eigen function?

✱(x)- State results of Stern-Gerlach experiment.

$L = \frac{1}{2} \hbar$

Q.2. Solve the following.

(3×10=30)

(i)- State and prove generalized uncertainty relation (Uncertainty principle for operators).

(ii)- Differential equation for simple harmonic oscillator obtained after solving Schrodinger wave equation is given by,

$$H''(z) - 2zH'(z) + (\lambda - 1)H(z) = 0$$

Solve this equation to find energy eigen values and wave functions of simple harmonic oscillator.

(iii)- Discuss the significance of ladder operators $\hat{L}_\pm$. Show that they have following property:

$$\hat{L}_+ \psi(l, m) = \sqrt{l(l+1) - m(m+1)} \hbar \psi(l, m+1)$$



UNIVERSITY OF THE PUNJAB

B.S. 4 Years Program / Sixth semester – Spring 2023

Paper: Quantum Mechanics-I

Course Code: PHYS-3301

Time: 3 Hrs. Marks: 60

THE ANSWERS MUST BE ATTEMPTED ON THE ANSWER SHEET PROVIDED

Q.1. Answer the following short questions.

(10x3=30)

- Prove that eigen value of Hermitian operator is real.
- Define wave function. What are properties of a wave function?
- Find out if the function $\Psi(x) = e^{ax}$ is an eigen function of x-component of momentum operator.
- Consider two states

$$|\Psi_1\rangle = 2i|\phi_1\rangle + |\phi_2\rangle - \alpha|\phi_3\rangle + 4|\phi_4\rangle$$

$$|\Psi_2\rangle = 3|\phi_1\rangle - i|\phi_2\rangle + 5|\phi_3\rangle - |\phi_4\rangle$$
 Where $|\phi_1\rangle, |\phi_2\rangle, |\phi_3\rangle$ and $|\phi_4\rangle$ are orthonormal kets and is α constant. Find the value of α , if $|\Psi_1\rangle$ and $|\Psi_2\rangle$ are orthogonal.
- What is physical significance of Hamiltonian in Quantum Mechanics?
- Define complementary and correspondence principles.
- Write down physical significance of ladder operators in Quantum mechanics.
- In case of orbital momentum operator, can $\hat{L}^2$ and $\hat{L}_z$ both be simultaneously measured? Explain the reason.
- Compute $[\hat{S}_z, \hat{S}_x] = ?$
- Prove That $\hat{a}|n\rangle = \sqrt{n}|n-1\rangle$. Where $\hat{a}$ is annihilation operator.

Answer the following questions.

- Q. 2. (a) Explain the Stern-Gerlach Experiment in detail. (6)
- (b) Given that in spherical coordinates $\hat{L}_z = -i\hbar \frac{\partial}{\partial \phi}$, determine whether or not $\sin m\phi$ and $e^{im\phi}$ are eigen functions of $\hat{L}_z$. (4)

Q. 3. Find the reflection and transmission coefficients R and T for 1 dimensional potential step for the case when $E > V_0$, where

$$V(x) = \begin{cases} 0, & \text{for } x < 0 \\ V_0, & \text{for } x \geq 0 \end{cases}$$

And also show that $R + T = 1$. (3+3+4)

Q. 4. Define the term central potential. Starting with the time independent Schrodinger's wave equation, obtain an expression of radial wave function. (2+8)





THE ANSWERS MUST BE ATTEMPTED ON THE ANSWER SHEET PROVIDED

Q.1. Write short answer of the following questions: (10x3=30)

- (i)- State any three postulates of quantum mechanics. ✓
- (ii)- Define Hilbert space and hence Fock space in terms of Hilbert space. State similarities and differences in both spaces.
- (iii)- What is meant by square integrable functions? Explain ket and bra notations (Dirac notations) briefly.
- (iv)- What is importance of simple harmonic oscillator? Give it classical theory.
- (v)- Write down Hamiltonian of simple harmonic oscillator. Define new operators,

$$P = \frac{\hat{p}}{\sqrt{m\hbar\omega}} \quad \text{and} \quad Q = \hat{x} \sqrt{\frac{m\omega}{\hbar}}$$

Show that Hamiltonian can be expressed as,

$$H = \frac{\hbar\omega}{2} (Q^2 + P^2)$$

- (vi)- A simple harmonic oscillator in its ground state has wave function,

$$\psi_0(x) = \frac{\sqrt{a}}{\pi^{1/4}} e^{-a^2 x^2/2}, \quad a^4 = \frac{4\pi^2 mk}{\hbar^2}$$

Given that $\xi = ax$, $\int_0^\infty e^{-\xi^2} d\xi = \frac{\sqrt{\pi}}{2}$ and $\int_0^1 e^{-\xi^2} d\xi = (0.84) \frac{\sqrt{\pi}}{2}$

Show that probability of finding the particle outside the classical region is 16%.

- (vii)- Consider the normalized wave function,

$$\psi = \sqrt{\frac{a}{\pi}} e^{-\frac{a}{2}(x-x_0)^2}$$

Find expectation value $\langle x \rangle$.

- (viii)- Given that in spherical polar coordinates $L_x = -i\hbar \frac{\partial}{\partial \phi}$, show that

$$[L_x, \cos\phi] = i\hbar \sin\phi$$

- (ix)- Define ladder operators of angular momentum. Evaluate $[L_x, L_+]$. ✓

- (x)- What is principle of Stern-Gerlach experiment.

Q.2. Solve the following.

(3x10=30)

- (i)- Given two operators $\hat{A}$ and $\hat{B}$ such that $[\hat{A}, \hat{B}] = i\hat{C}$, then prove that the uncertainties $\Delta\hat{A}$ and $\Delta\hat{B}$ in an arbitrary state are related by,

$$\Delta\hat{A}\Delta\hat{B} \geq \frac{1}{2} |\langle \hat{C} \rangle|$$

- (ii)- Solve the Schrodinger wave equation for simple harmonic oscillator and show that it reduces to Hermite differential equation.

- (iii)- What is relation of spherical harmonics with orbital angular momentum? Find eigen functions of angular momentum operator.



THE ANSWERS MUST BE ATTEMPTED ON THE ANSWER SHEET PROVIDED

Q.1. Write short answer of the following questions:

(10x3=30)

i. What is physical significance of a wave function? Some wave functions are given below:

$$\psi_1(x) = x^2, \psi_2(x) = e^{ix}$$

Are these functions physically acceptable or not?

ii. Show that $\psi(x) = e^{-\frac{x^2}{2}}$ is eigen function of $(\frac{\partial^2}{\partial x^2} - x^2)$ and find eigen value.

iii. Check whether momentum operator $\hat{p}_x$ and position operator $\hat{x}$ commute? What conclusion you draw from result?

iv. What is physical significance of Hamiltonian in Quantum Mechanics?

v. Define complementary and correspondence principles.

vi. Write down any three postulates of Quantum Mechanics.

vii. Derive the equation of continuity for Schrodinger wave equation.

viii. Given that in spherical coordinates $\hat{L}_z = -i\hbar \frac{\partial}{\partial \phi}$, show that $[\hat{L}_z, \cos\phi] = i\hbar \sin\phi$

ix. Name the three pairs of canonically conjugate quantities.

x. Prove That $\hat{H} \hat{a}^\dagger |n\rangle = (E_n + \hbar\omega) \hat{a}^\dagger |n\rangle$. Where $\hat{a}^\dagger$ is the creation operator.

Solve the following.

Q. 2. Consider a system whose quantum mechanical wave function is given by

$$\psi = \frac{\sqrt{3}}{3} \psi_1 + \frac{2}{3} \psi_2 + \frac{\sqrt{2}}{3} \psi_3$$

Where ψ_1, ψ_2, ψ_3 are all orthonormal wave functions

a) Show that wave function is normalized. (2)

b) Calculate the probability of finding the system is each of the states ψ_1, ψ_2, ψ_3 . (6)

c) What is value of total probability? (2)

Q.3. Solve Schrodinger wave equation for a particle in one dimensional box. Also find energy eigen values and normalized wave function. (3+3+4)

Q.4. (a) State and prove Ehrenfest's Theorem. (6)

(b) Prove that eigen value of Hermitian operator is real. (4)



Q.1. Give short answers to the following questions.

(10x3=30)

- How do you define term "Wave Function" in quantum mechanics. Write down its physical significance.
- Show that Hamiltonian operator and momentum operator commutes with each other.
- Explain any three postulates of quantum mechanics.
- Show that ψ is an eigen function of an operator and determine eigen value if
- Write down physical significance of position momentum uncertainty.
- What is zero-point energy? Calculate zero-point energy of one-dimensional harmonic oscillator.
- Consider two states
Find the value of 'a' so that ψ_1 and ψ_2 are orthogonal.
- State correspondence principle.
- Show that
- Define ladder operator in quantum mechanics. Give their importance in any physical phenomenon.

Answer the following questions in detail.

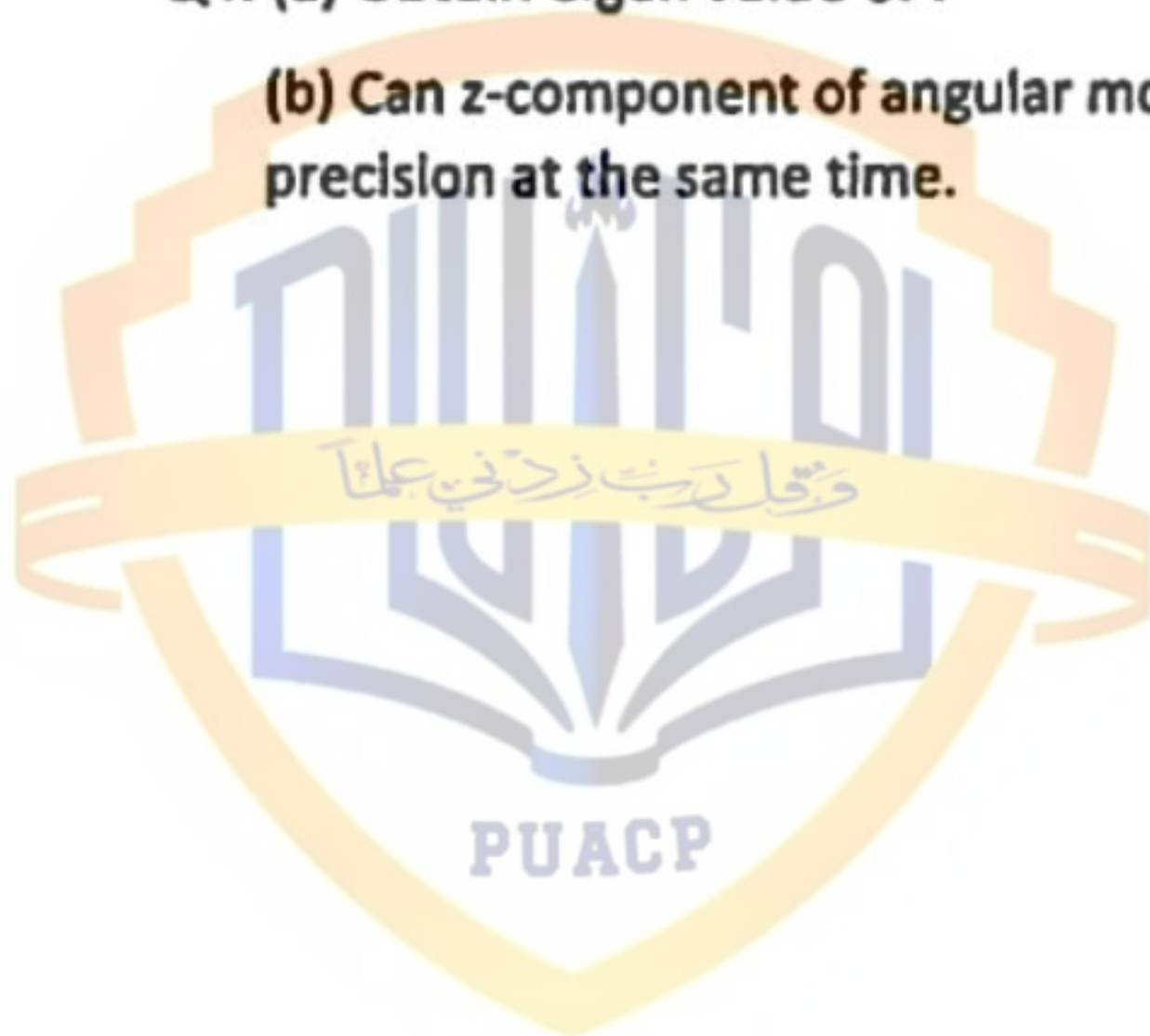
Q2. (a). Derive equation of continuity and explain its physical significance in hydrodynamics system.

**(b). If ψ is normalized wave function. Find the expectation value of x where $0 < x \leq 10$.
(5+5)**

**Q3. A particle of mass m and total energy $E > V_0$ strikes a potential step from left where
Calculate reflection and transmission coefficient. Also show that $R+T=1$ (10)**

Q4. (a) Obtain eigen value of L_z .

(b) Can z-component of angular momentum and azimuthal angle can be measured with precision at the same time. (6+4)





UNIVERSITY OF THE PUNJAB
B.S. 4 Years Program : Fifth Semester – 2020

Paper: Quantum Mechanics-I
Course Code: PHY-305

Part – II

Roll No. [REDACTED]
Time: 2 Hrs. 45 Min. Marks: 50

ATTEMPT THIS (SUBJECTIVE) ON THE SEPARATE ANSWER SHEET PROVIDED

Q.2. Give short answers of the following:

(5×4=20)

- i) Differentiate between observable and operator?
- ii) Write the properties of an adjoint operator.
- iii) Define complementarity and correspondence principles.
- iv) Define a) uncertainty relation b) expectation value.
- v) Define quantum mechanical tunneling.

Answers the following questions.

(3×10=30)

Question No. 3

A beam of electrons having energy E is incident on a step potential of height V_0 . Find reflection and transmission coefficients when $E > V_0$. Show that $R + T = 1$.

(10)

Question No. 4 ✓

a) Solve schrodinger's wave equation for a particle in one dimensional box.

(5)

Find energy eigen values and normalized wave function.

(5)

Question No. 5

a) What is angular momentum. Define operator for angular momentum. Also find commutation relation for y and z components of angular momentum.

(5)

b) Write down postulates of Quantum Mechanics.

(5)





UNIVERSITY OF THE PUNJAB

Fifth Semester – 2019

Examination: B.S. 4 Years Program

Roll No.

PAPER: Quantum Mechanics-I

Course Code: PHY-305 Part – II

MAX. TIME: 2 Hrs. 45 Min.

MAX. MARKS: 50

ATTEMPT THIS (SUBJECTIVE) ON THE SEPARATE ANSWER SHEET PROVIDED

Q2: Give short answers to the following questions:

(4x5 = 20)

- Define a) expectation value b) eigen value of an operator
- What is wave mechanics?
- Explain degenerate spectrum.
- What is importance of angular momentum in quantum mechanics?
- What is essence of correspondence principle?

Q3: (a) Show that eigen functions corresponding to different eigen values are orthogonal.

(b) Define Skew operator and hermitian operator.

(7+3)

Q4: (a) Derive time independent Schrodinger's wave equation for a single particle.

(b) Write down three basic postulates of Quantum Mechanics.

(7+3)

Q5: Find reflection coefficient and transmission coefficient of a beam of particles with energy greater than height of step potential.

(10)



UNIVERSITY OF THE PUNJAB

Fifth Semester 2018
Examination: B.S. 4 Years Programme

Roll No. 014271

PAPER: Quantum Mechanics-I
Course Code: PHY-305

TIME ALLOWED: 2 hrs. & 30 mins.
MAX. MARKS: 50

Attempt this Paper on Separate Answer Sheet provided.

SUBJECTIVE TYPE

i.

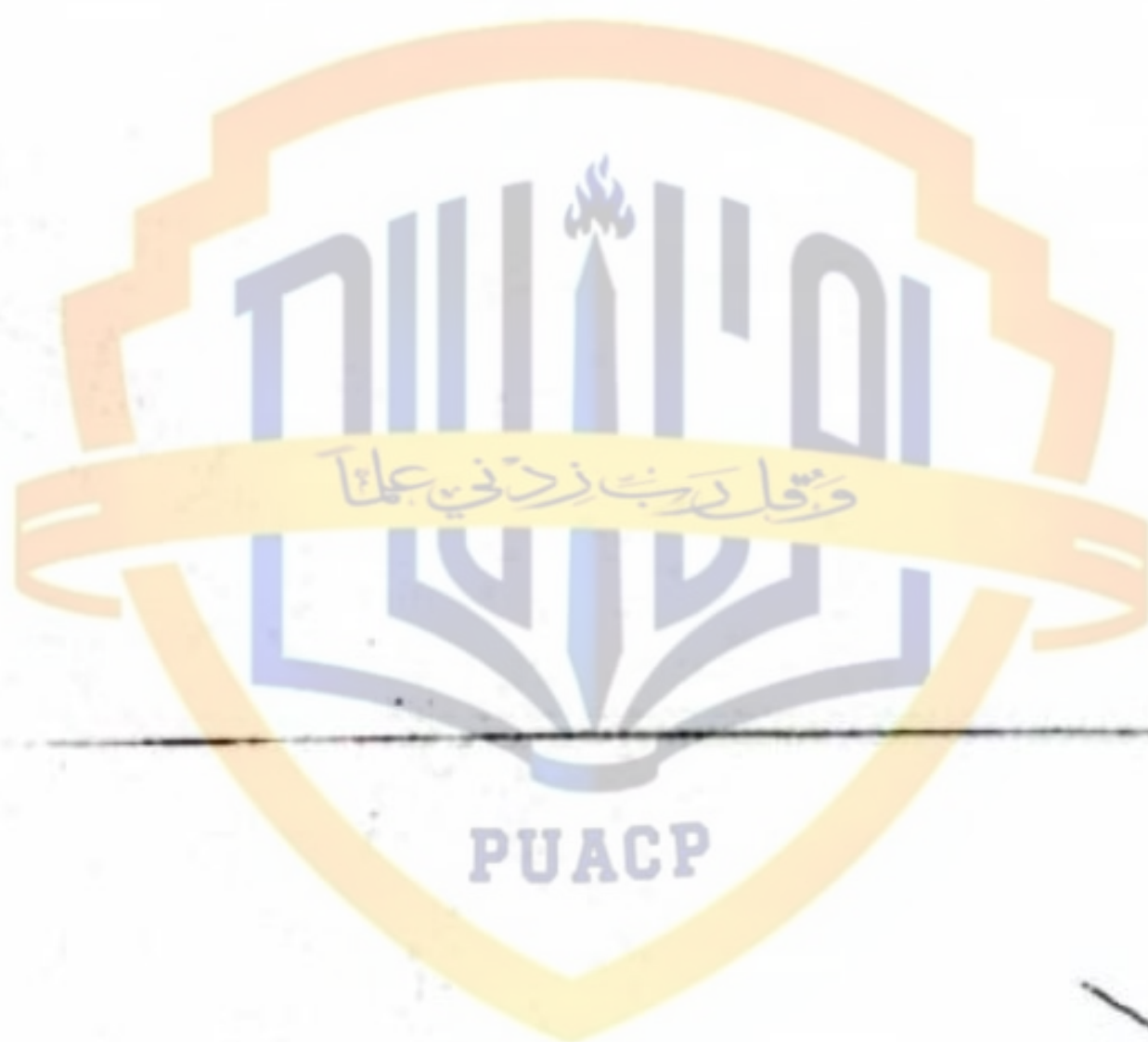
Q2: Give short answers to the following questions: (4x5 = 20)

- What is zero point energy, If a classical oscillator has energy $\frac{1}{2} h \omega$, What is its amplitude? $E_{\text{cm}} = \frac{1}{2} m \omega^2 A^2$
- Define degenerate eigen values, non-degenerate eigen values, linear dependent functions and linear independent functions.
- Describe Correspondence principle.
- State Hilbert space and give two of its examples.
- Write physical significance of Uncertainty principle.

Q3: Define the term Central potential. Starting with the time independent Schrodinger's wave equation, obtain an expression of radial wave function. (10)

Q4: (a) If two operators have simultaneous eigen function, then these operators commute
(b) Write down three postulates of Quantum Mechanics. (7+3)

Q5: (a) Find eigen value and eigen function of z-component of angular momentum.
(b) Prove that $[\hat{L}_z, \sin\phi] = -i\hbar \cos\phi$ (7+3)





UNIVERSITY OF THE PUNJAB

Fifth Semester 2017

Examination: B.S. 4 Years Programme

Roll No. 12.12.17

PAPER: Quantum Mechanics-I
Course Code: PHY-305

TIME ALLOWED: 2 hrs. & 30 mins.
MAX. MARKS: 50

Attempt this Paper on Separate Answer Sheet provided.

SUBJECTIVE TYPE

Q.2 Give short answers to the following questions. (4x5=20)

- Define Hermitian operator and prove that $\hat{x}$ is hermitian.
- Show that the Eigen value spectrum of $\hat{L}_x$ is $m\hbar$. Where m has integer values.
- Describe the terms briefly, complete set of linear independent objects, degenerate Eigen values.
- Briefly explain the physical interpretation of ladder operator in Quantum mechanics.
- What is a zero point energy? If a classical oscillator has energy $\frac{1}{2}\hbar\omega$, what is its amplitude?

Q.3(a) State and prove uncertainty principle in Quantum mechanics.

(b) Show that $(\Delta A)^2$ is always real number. (6+4)

Q.4 (a) Write down the expressions for the angular momentum operators $\hat{L}_x, \hat{L}_y, \hat{L}_z$ in Cartesian coordinates and convert them into spherical polar coordinates.

(b) Also calculate eigen value spectrum of $\hat{L}_z$. (6+4)

Q.5 (a) State and prove Ehrenfest's Theorem. (2+5,3)

(b) Write down three canonical conjugate observable.



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UNIVERSITY OF THE PUNJAB

Fifth Semester 2016
Examination: B.S. 4 Years Programme

Roll No. 009583

PAPER: Quantum Mechanics-I
Course Code: PHY-305

TIME ALLOWED: 2 hrs. & 30 mins.
MAX. MARKS: 50

Attempt this Paper on Separate Answer Sheet provided.

SUBJECTIVE TYPE

Q.2. Write short answers of the following questions (4x5=20)

1. Write down the statement of any two postulates of quantum mechanics.
2. Write down the expression for the operators for angular momentum.
3. What is expectation value? Write down its expression.
4. Discuss the superposition principle.
5. Define Hermitian operator. Give an example.

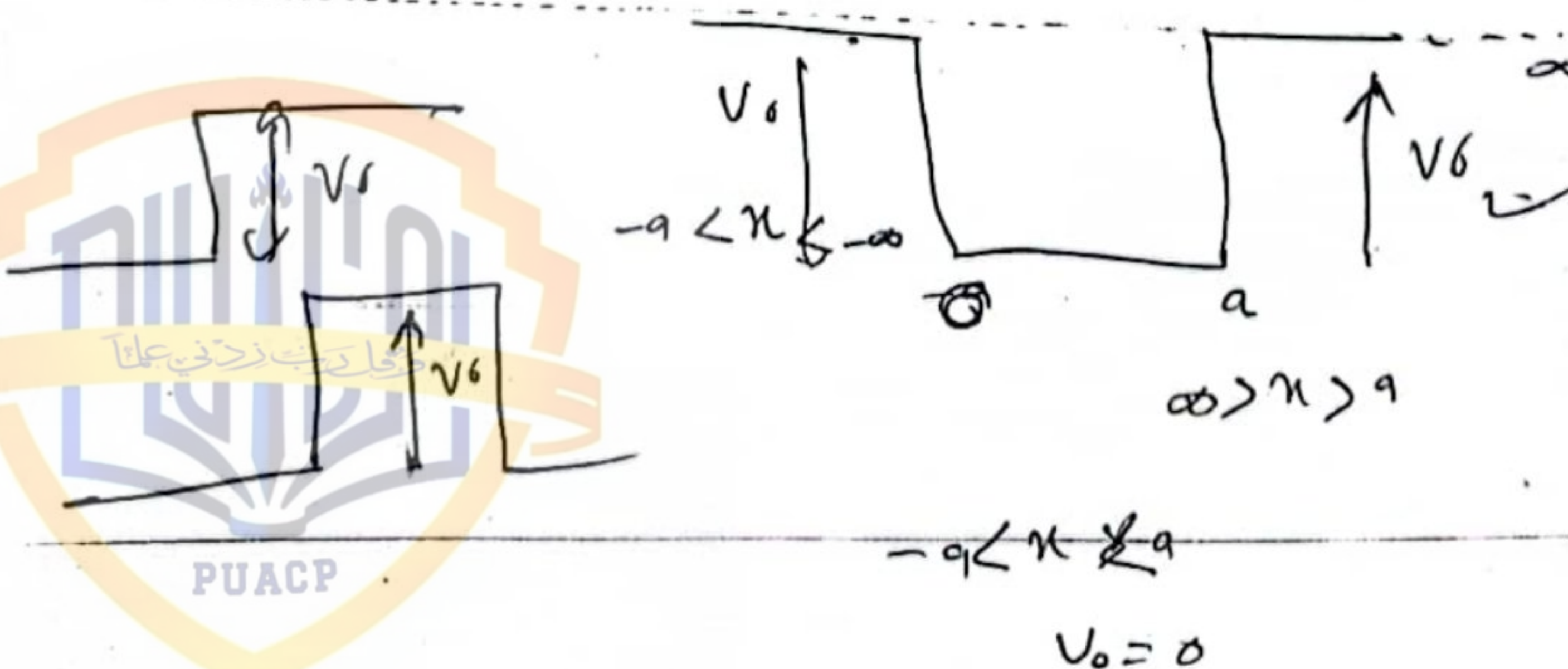
Q.3. Explain briefly

(3x10=30)

1. Derive equation of continuity in quantum mechanics. (10)
2. State and prove uncertainty principle in quantum mechanics. (10)

3. A particle enters a potential well such that

$V=0$ when $x < 0$ and $V=V_0$ for $0 < x < a$ and $V=0$ for $x > a$. Solve time independent Schrödinger wave equation and show that $R+T=1$. (10)



Attempt this Paper on Separate Answer Sheet provided.

SUBJECTIVE TYPE

(4x5=20)

Q.2 Give short answers to the following questions.

i. Explain the term "Zero point" energy of simple harmonic oscillator. — 3.22

ii. Briefly explain the significance of Ladder operators in Quantum mechanics.

3.31 — iii. State and explain principle of superposition in quantum mechanics.

iv. If a classical oscillator has energy $\frac{1}{2} \hbar \omega$, what is its amplitude?

v. Using generalised uncertainty principle $\Delta A \Delta B \geq \frac{1}{2} |\langle [\hat{A}, \hat{B}] \rangle|$

calculate uncertainty relation between components of position x and p_x

2.1/23 — Prove that the eigenfunctions of a hermitian operator belonging to distinct eigenvalues are orthogonal. (10)

Q.4 Write down the expressions for the angular momentum operators $\hat{L}_x, \hat{L}_y, \hat{L}_z$ in Cartesian coordinates and convert them in spherical polar coordinates. Also calculate the eigenvalue spectrum of $\hat{L}_z$. (6+4)

Q.5 Let $\hat{A}$ and $\hat{B}$ be two operators and let ΔA and ΔB be uncertainties in the measurement of $\hat{A}$ and $\hat{B}$ respectively. Show that $(\Delta A)^2 (\Delta B)^2 \geq \left| \frac{1}{2i} \langle [\hat{A}, \hat{B}] \rangle \right|^2$ (10)

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UNIVERSITY OF THE PUNJAB

Fifth Semester 2013
Examination: B.S. 4 Years Programme

Roll No. ...2681...

PAPER: Quantum Mechanics-I
Course Code: PHY-305

TIME ALLOWED: 2 hrs. & 30 mins.
MAX. MARKS: 50

Attempt this Paper on Separate Answer Sheet provided.

SECTION II (Subjective)

2. Give short answers to the following questions

(4×5=20)

i. Show that $(\Delta A)^2$ is always non negative real number

ii. Briefly explain the significance of Ladder operators in Quantum mechanics

iii. What is meant by Degenerate eigen values? Explain with the help of an example

iv. If a classical oscillator has energy $\frac{1}{2}\hbar\omega$, what is its amplitude?

v. Define the terms Hermitian operator, Wave front, Plane wave, Wave packet.

vi. What is meant by Eigen value spectrum? Find out the eigen value spectrum of Hamiltonian $\hat{H}$ for simple harmonic oscillator.

(2+3)

vii. Write down the expressions for the angular momentum operators $\hat{L}_x, \hat{L}_y, \hat{L}_z$ in cartesian coordinates and convert them in spherical polar coordinates. Also calculate the eigen value spectrum of $\hat{L}_x$.

(6+4)

Repeat

viii. Define the term central potential. Starting with the time independent Schrodinger wave equation, obtain the radial wave equation for central potentials.

(10)

